



Pushparaj E

 pushparajraje52141@gmail.com  +91 7603881811

 Chennai , Tamil Nadu  linkedin.com/in/pushparajraje

PROFESSIONAL SUMMARY

Results-driven Software Engineer with **3 years of experience in full-stack development, enterprise ERP systems, and AI-powered automation**. Proven success building scalable CRM and ERP platforms that streamline operations for **65+ internal users**. Specialized in **JavaScript, Node.js, React, and PostgreSQL**, with hands-on expertise in predictive maintenance systems and document automation using LLM technologies. Built and scaled CRM systems handling **4,900+ leads**, transforming manual workflows into data-driven software solutions that **reduce operational overhead by 40%** and improve cross-functional efficiency.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), Python, SQL, HTML5, CSS3

Frameworks & Libraries: Node.js, Express.js, React.js, LangChain, Next.js

Databases: PostgreSQL, MongoDB, Firebase Realtime Database

DevOps & Tools: Git, Postman, Jira, Figma, CI/CD, Render, Linode

AI/ML Technologies: LLM Integration, Ollama, Groq, AI Agents, Predictive Analytics, Anomaly Detection

PROFESSIONAL EXPERIENCE

Software Engineer – AI Systems & Full Stack

Jul 2025 – Present

Petrokens Engineering and Services Pvt Ltd
Chennai, India

1. ProjectEPC – Enterprise ERP Platform - (Production)

project overview: Enterprise-grade ERP platform designed to streamline operations and improve cross-departmental visibility for a mid-sized engineering organization.

- Designed and deployed a comprehensive ERP system serving **65+ employees**, reducing operational bottlenecks by **45%**.
- Built scalable backend architecture with **500+ REST APIs** and **130+ PostgreSQL tables**, enabling real-time cross-department data flow.
- Optimized database schema, eliminating **60% data redundancy** and improving reporting performance by **3x**.
- Implemented **RBAC with audit logging** and automated document lifecycle workflows, achieving **zero unauthorized access incidents** and reducing manual coordination.

2. AI-Powered Pump Predictive Maintenance System - (In Development)

project overview: AI-driven predictive maintenance system designed to detect early equipment failures using industrial sensor data.

- Developing machine learning-based predictive maintenance solution analyzing vibration, temperature, and pressure sensor data for early anomaly detection
- Processing **10,000+ daily sensor readings** through anomaly detection algorithms to support proactive maintenance scheduling.

3. AI Document Intelligence for QA/QC (R&D Prototype)

project overview: LLM-powered document intelligence system to automate quality assurance and technical document review processes.

- Building **LLM-powered document analysis system** using **LangChain**, **Ollama**, and **Groq** to support automated technical document review workflows
- Architecting contextual memory and intelligent summarization pipelines for extracting insights from **100+ page engineering documents**

Full Stack Developer & Trainer

UrbanCode Edutech Pvt Ltd
Chennai, India

Apr 2024 – Jul 2025

1. Zen Lead Management System

project overview: Zen Lead Management System is an enterprise- grade CRM designed to manage the complete student lifecycle, handling 4,900+ leads with high performance and scalability.

- Architected enterprise-grade CRM platform managing the complete student lifecycle, processing **4,900+ leads**
- Developed **50+ secure REST APIs** with JWT authentication and RBAC, improving API response time to **<200ms**
- Designed scalable PostgreSQL database schema supporting **50,000+ records** with complex query execution in **<50ms**
- Implemented automated communication workflows and dashboards, reducing manual tasks by **40%** and improving lead response time by **65%**

Full Stack Developer

Leather Sector Skill Council (LSSC)
Chennai, India

Jun 2023 – May 2024

1. Scale LMS Platform

project overview: Scale LMS Platform is a comprehensive Learning Management System built to support 500+ learners, delivering an engaging learning experience with an 85% course completion rate.

- Built comprehensive Learning Management System serving **500+ learners**, achieving **85% course completion rate**
- Engineered real-time progress tracking dashboards with Firebase integration, reducing administrative overhead by **35%**
- Optimized platform performance through code splitting and lazy loading, improving page load times by **60%** and enhancing learner engagement by **45%**

ACHIEVEMENTS

- Trained **50+ students** in MERN stack, Python, and Java through mentoring and technical training
- Built **3+ production LMS and CRM platforms** for real-world organizations
- Delivered **5+ client websites**, improving their digital presence and online reach
- Developed enterprise ERP modules supporting operations for **65+ users**
- Contributed to AI-based automation systems for predictive maintenance and document intelligence

EDUCATION

B.E.Computer Science Engineering At Kings Engineering College | CGPA - 8.5

SOFT SKILLS

Team Collaboration | Problem Solving | Time Management | Mentoring & Leadership | Communication